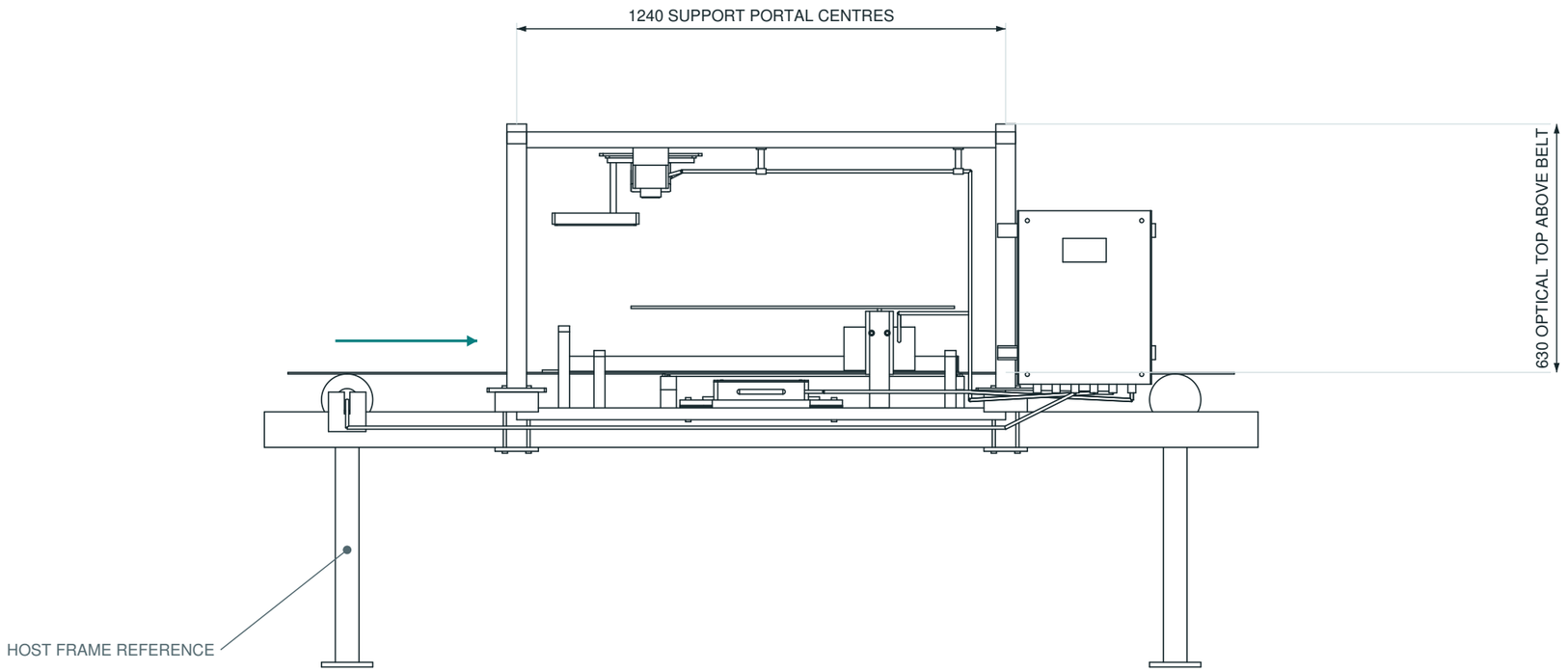
Retrofit sensing module / general arrangement

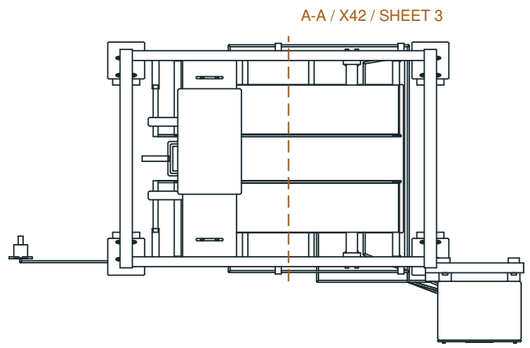
The module is the proposed product. The existing recycling line is host context only.



SIDE / installed reference assembly / 1:15

HOST BELT TOP: Z1000 / TRAVEL +X

A-A / X42 / SHEET 3



PLAN / module only, existing belt omitted / 1:25

PRODUCT BOUNDARY

Sensor head, optical hardware, two-direction electromagnetic research cartridges, mounting adapters, encoder association, acquisition/control enclosure and supported cabling.

EXISTING LINE

Host belt, supporting deck, frame and roller references are not part of the proposed product. Integration is conditional on host compatibility.

A CONTROLLED EXPERIMENT

Only a central 180 mm lane is constrained. The outer host lanes are unused. Neither nominal lane width nor modeled coil size establishes useful sensing coverage.

COORDINATE BASIS

X follows travel, Y is across the belt, Z is up. All positions are proposed coordination dimensions, not approved installation dimensions.

SEPARATE VALIDATION

The CAD demonstrates a packaging hypothesis. Test magnetic background, dynamic association, classifier value, cleaning, safety and maintainability before an industrial pilot.

DIMENSIONS: mm | SCALE PER VIEW AT A3 | PROPOSED RESEARCH GEOMETRY

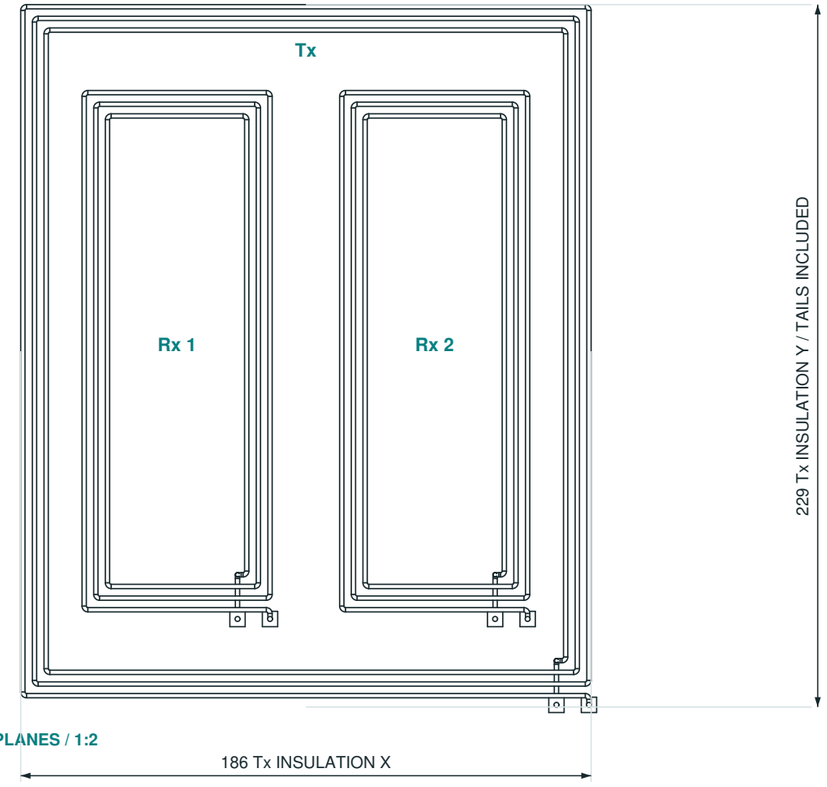
No approved fit, tolerance, load, electromagnetic coverage, throughput or safeguarding rating.

Source: module_with_host.step | SHA-256 5f091afbdf702875 | geometry revision M01-W1

Two complementary excitation directions

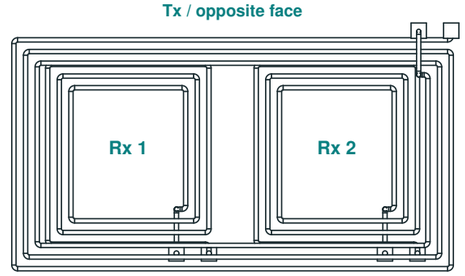
Six functional winding paths: 2 Tx and 4 Rx. Three illustrative perimeter passes per path; unoptimized geometry.

Excitation axis principally Z (out of this view)

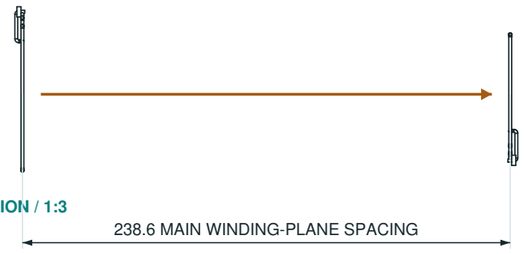


A / UNDER-BELT / XY COIL PLANES / 1:2

Excitation axis principally Y (across the lane)



B / LOOKING ACROSS Y AT XZ COIL PLANES / 1:2



B / END / SIDE-TO-SIDE EXCITATION / 1:3

VIEW GEOMETRY: INSULATED PATHS + TERMINAL PINS / BASES; FORMERS OMITTED.

A: ONE Tx + TWO PROPOSED Rx

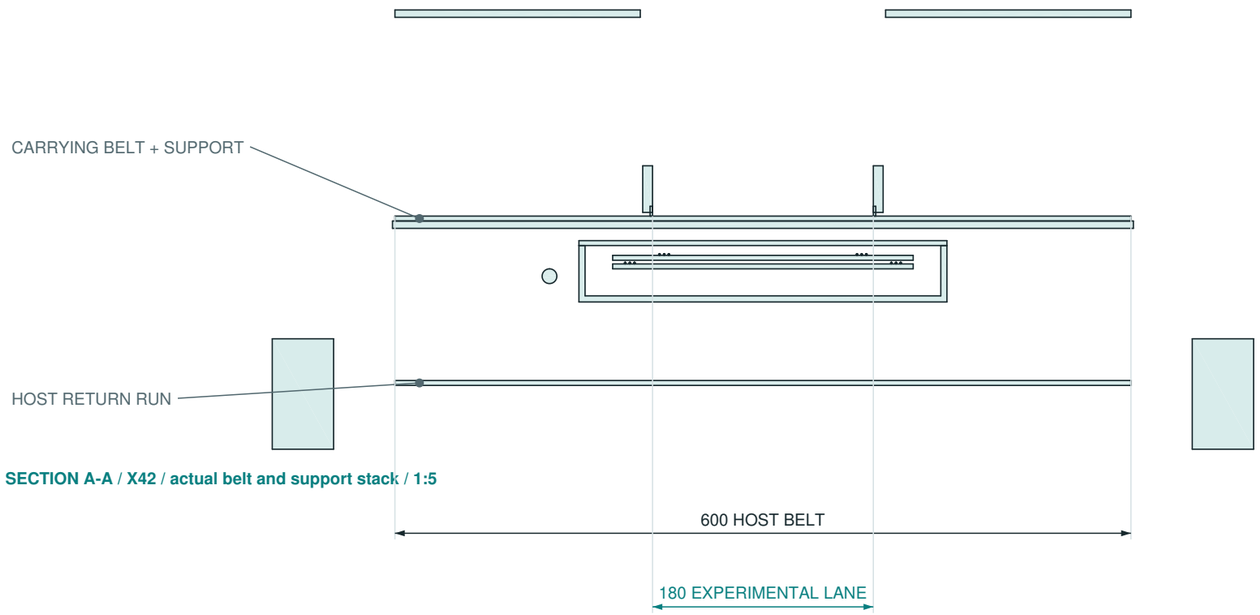
Illustrative core diameter 0.9, insulation OD 1.4, pitch 3.8. Belt-top gaps Tx/Rx: core 37.85/30.85; insulation 37.6/30.6; terminal pins 37.2/30.2. Returns occupy cut former grooves; retention and manufacture remain unselected.

B: ONE Tx + TWO PROPOSED Rx

Vertical XZ paths provide a complementary transverse axis. Open spiral transitions are not three exact closed turns. Individual start/finish pins do not define acquisition wiring, polarity or differential pairing. Frequency, balance and useful fields remain unselected.

Host interface and experimental exclusion reservation

A nonmetallic window alone does not establish a compatible installation.



SECTION A-A / X42 / actual belt and support stack / 1:5



DETAIL / X42 / stack dimensions listed at right / 1:2

Stack item	mm
Carrying belt	4
Support window	6
Head-to-support gap	10
Rx core-top gap	30.85
Rx insulation-top gap	30.6
Rx pin-top gap	30.2
Return top to shell	64

Loaded sag, tolerances and conductive-feature height add uncertainty. No deflection allowance or usable lift-off is validated.

MEASURE THE BACKGROUND

The dashed scene reservation is X -180..480, Y -340..340, Z850..1200. This 660 x 680 x 350 mm box is a test allowance, not a verified electromagnetic exclusion distance.

HOST ACCEPTANCE

Measure the actual belt reinforcement, joint, supports, rollers, frame, fasteners and contamination across the intended frequency range. Retest under load and motion; quantify drift and background cancellation.

PROPOSED ADAPTERS

Support portal centres: X +/-620, Y +/-420. Slots: 24 x 9. Host hole pattern, selected fasteners, load paths and permissible deflection remain open.

LANE CONTROL

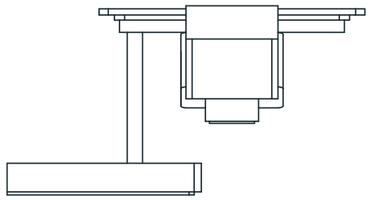
Upstream masks, compliant skirts and proposed pod guards restrict outer-lane access. Feed metering must inhibit blocked or off-lane material. Jams, wear, escape and impact remain untested; no accepted fragment-size limit is implied.

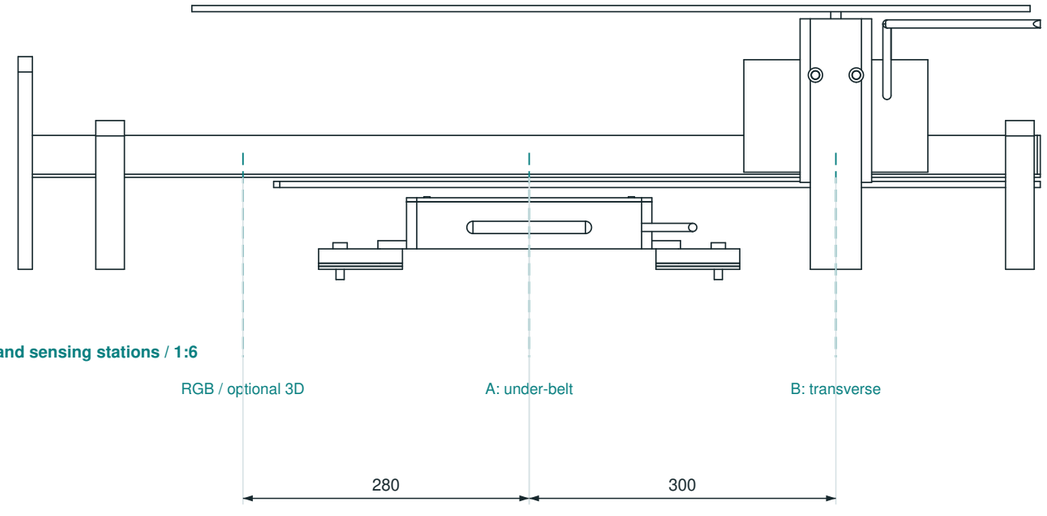
INTEGRATION HOLD

Metal-free packaging is a design intention, not proof of sensing performance. Do not install on an unknown reinforced belt or assume full-width coverage.

Optical observation and fragment association

RGB + optional exterior 3D geometry condition the interpretation of directional response sequences.





SIDE / optical head, lane and sensing stations / 1:6

VISIBLE SENSOR PACKAGE

Adjustable RGB head and illumination, plus an optional exterior 3D geometry device. Lens, resolution, exposure, optical protection, calibration and dust management remain to be selected.

Proposed datum	mm
RGB centre XYZ	-280 / 0 / 1500
Lens barrel datum Z	1445
Optional 3D XYZ	-280 / 150 / 1495
Footprint example	260 x 180

GEOMETRY, NOT HIDDEN CONTENT

The optional 3D device measures exterior shape/pose. No optical device is shown measuring buried copper, and no electromagnetic response is treated as a validated compositional assay.

RESEARCH HYPOTHESIS

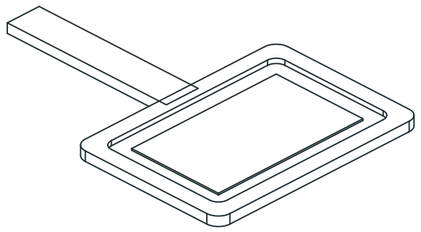
Pose/geometry-conditioned directional and frequency responses may help distinguish conductive structures in layered fragments. Compare against strong simpler baselines; generic optical/induction fusion already exists.

MOVING ASSOCIATION

Host encoder reference X-1050. Station observations are sequential. Last required usable observation is TBD; downstream timing starts there. Slip, overlap, rolling and pose changes need explicit handling.

Service and calibration access

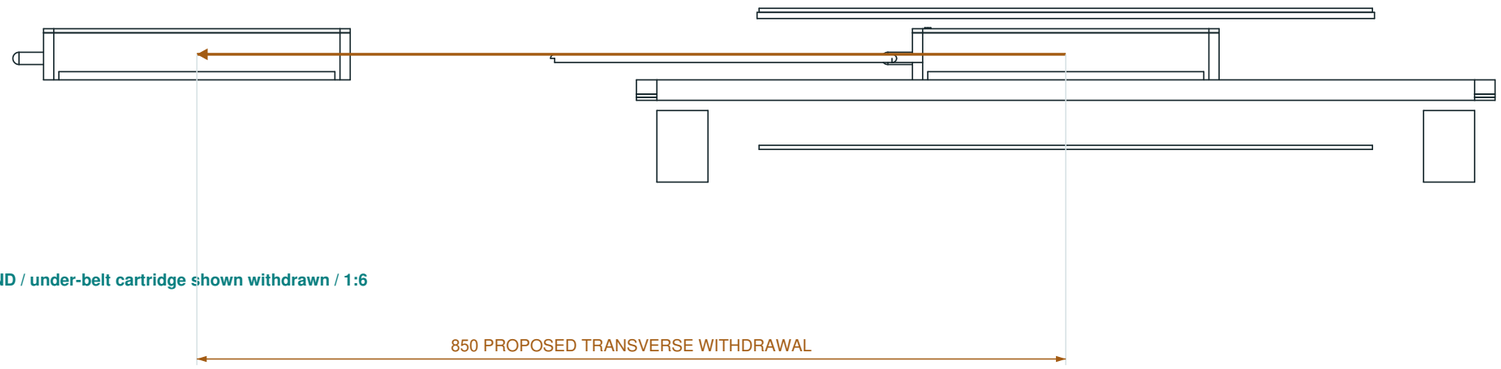
Nominal and removed views use the same source solids; no service or isolation procedure is certified.



REMOVABLE CALIBRATION ACCESSORY / CAD SOLIDS / 1:3

REFERENCE ACCESSORY

Nonmetallic carrier and handle with an unselected coupon envelope. Illustrative carrier: 100 x 140 mm. Material, target response, repeatability and safe insertion remain unresolved.



END / under-belt cartridge shown withdrawn / 1:6

CALIBRATION INSERT

Reserve a repeatable target insertion path along -X, 450 mm proposed. Define traceable references for pose, conductivity, geometry, temperature and frequency response before claiming calibrated sensing.

OPTICAL ACCESS

Proposed lid removal: +200 mm Z. Allow lens cleaning and camera adjustment, then restore calibration and enclosure integrity. No optical device rating or ingress protection is selected.

CONTROL ENCLOSURE

Front panel withdrawal: 350 mm toward -Y. Maintain cable slack, connector access and a protected work area. Power isolation, interlocks and protective bonding require separate design.

GEOMETRIC CHECK SCOPE

Exact CAD solids are shown at nominal and proposed service offsets. Read the separate verification report for the swept-interference check and exclusions. Cable reach, hand/tool access and safe servicing remain unvalidated.

WHAT REMAINS OPEN

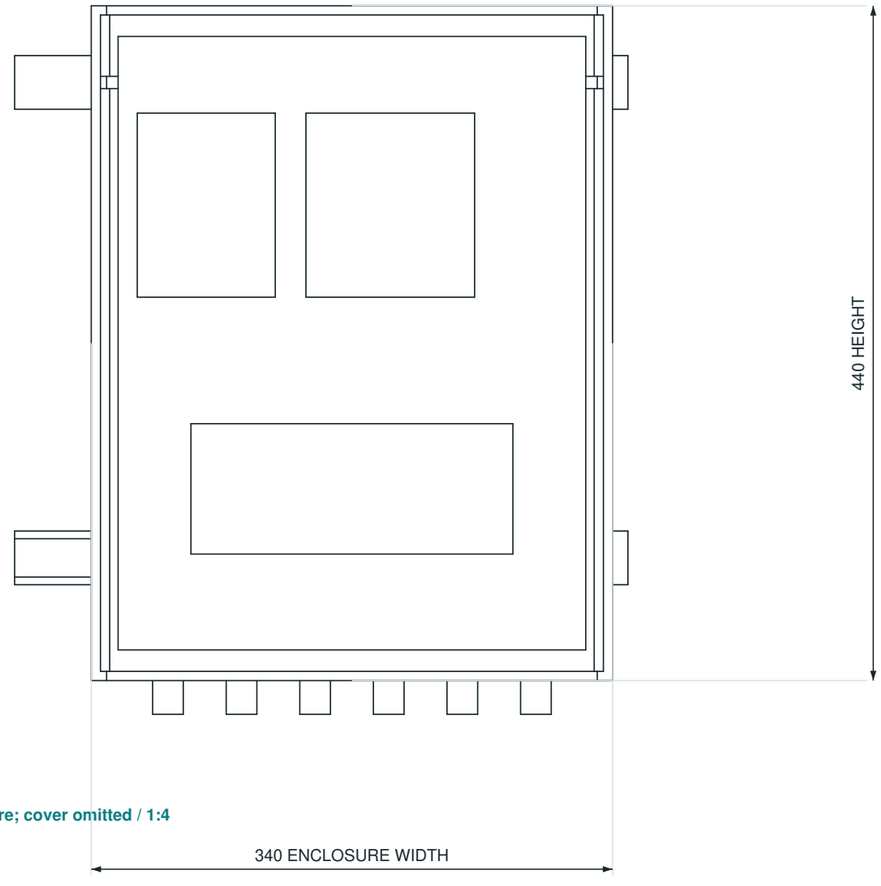
Real tools, hands, lift/support devices, connector pull lengths, sag, wear and operator access are not validated. Component selection and maintainability trials precede production detailing.

CLEARANCE IS CONDITIONAL

Isolate the line, release access hardware, disconnect and support the cartridge, then withdraw toward -Y. Coordinate the service aisle with the existing line and route all removable leads for disconnection. An empty translation envelope is not a safe servicing method.

Acquisition / control and interface package

Mechanical envelopes only. No selected electronics, safety architecture or released circuit design.



FRONT / actual enclosure; cover omitted / 1:4

Interface	Proposed responsibility
RGB / optional 3D	Observation + geometry/pose
Coil Tx / Rx leads	Excitation and response acquisition
Encoder input	Travel / timestamp association
Module software	Alignment, confidence, diagnostics
Host line interface	Decision / data integration only

HOST ACTUATION IS OUTSIDE THIS MODULE

Ejectors, air supply, diverters, bins and conveyor drive are not supplied by this concept. Any future command interface needs agreed timing, fail-safe behavior and a line-level safety review.

ELECTRONIC DESIGN HOLDS

Select excitation frequencies, power, low-noise receivers, digitizers, timing, shielding, grounding, isolation and EMC strategy. Establish temperature stability, calibration and self-test before defining the enclosure and harness.

EVIDENCE BOUNDARY

Existing computer-vision annotation-recall results are separate from this unbuilt sensing module. They are not physical copper recovery, purity, payback or proof of a benefit from multi-angle sensing.

SERVICEABLE PACKAGING

Separate acquisition/control enclosure, internal component envelopes, removable cover, connector access and supported cable routes. The modeled volume does not verify heat rejection or EMC.

DIMENSIONS: mm | SCALE PER VIEW AT A3 | PROPOSED RESEARCH GEOMETRY

No approved fit, tolerance, load, electromagnetic coverage, throughput or safeguarding rating.

Source: module_with_host.step | SHA-256 5f091afbdf702875 | geometry revision M01-W1